EduEats

Minor Project-II

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Submitted in partial fulfilment of the requirement of the degree of

BACHELOR OF TECHNOLOGY

to

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by

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CERTIFICATE

This is to certify that the Project Synopsis entitled, "**EduEats**" submitted by "**Adarsh Kumar Sharma(2301420058) and Shivam Raj(2301420057)"** to **K.R Mangalam University, Gurugram, India,** is a record of bonafide project work carried out by them under my supervision and guidance and is worthy of consideration for the partial fulfilment of the degree of **Bachelor of Technology** in **Computer Science and Engineering** of the University.

Type of Project (Tick One Option)

Industry/Research/University Problem

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ABSTRACT

In today's fast-paced college environment, students often struggle with long queues and delays while ordering food from campus cafeterias. The EduEats project aims to provide a smart, efficient, and user-friendly food ordering system specifically designed for college students and staff. The system allows users to place food orders from their mobile devices or computers, reducing waiting time and ensuring timely delivery.

The integration of a digital menu, order tracking, and payment options creates a seamless experience that enhances user satisfaction. With the growing demand for digital solutions in every sector, a smart food ordering system can improve convenience, operational efficiency, and reduce food wastage through real-time tracking.

Utilizing technologies such as Mongo, Express.js, React, and Node.js, EduEats enables cafeteria owners to manage orders efficiently while offering students a hassle-free ordering experience.

The platform ensures a user-friendly interface, secure payment integration, and a real-time order tracking system, enabling a hassle-free experience. EduEats also benefits cafeteria vendors by streamlining operations, reducing congestion, and improving customer satisfaction. The system leverages modern web development technologies, ensuring responsiveness and accessibility across devices.

KEYWORDS: Food Ordering, Campus, Digital Menu, Order Tracking, Smart System

Chapter 1
Introduction

1. Background of the project

Urbanization, rapid economic liberalization, growing large-scale political turmoil, fierce conflicts, and inadequate and inappropriate policies are the basis of crime in urban areas. In addition, poverty and inequality caused by rising expectations and a sense of moral anger among some members of society contribute to increasing and increasing the level of crime. Structural adjustment programs pursued to promote economic growth, such as layoffs, shrinking civil servants, and selling civil servants, have led to increased poverty and inequality. One of the consequences of these programs is the sharp rise in unemployment, which is a major cause of crime epidemics and increases, especially in urban areas.

Not just this, Covid-19 pandemic and subsequent lockdown increased the crime rates in India itself by 28%. Unfortunately, the world we live in is becoming increasingly unsafe for all of us. We've seen a steady increase in the registered crime cases over the decade. Using Home Surveillance Camera we can detect crimes such as kidnapping, murder, burglary, etc. The emergence of closed-circuit television (CCTV) surveillance is a mainstream crime prevention measure used around the world. CCTV can dramatically increase the security of your property and keep your family protected. Once you get CCTV installed in your house, you no more need to worry about break-ins or burglary while you are away. Because whatever happens, it will still be recorded in your CCTV cameras. Thus, these cameras let you be, free of worry while you are away on a vacation or especially during business trips. It can very well help mitigate the crime in the first place and curb it. If a potential burglar sees your household is fully secured with CCTV in every nook and corner, it might as well put him off and because of the fear, he will deter from any crime that he earlier intended to do.

The importance of adequate security measures for homes and business spaces cannot be stressed enough. Many shops, commercial spaces, educational institutions, and public areas are now under the watchful eyes of surveillance systems. The recordings can be used to monitor and thwart crime. CCTV footages can also be produced as evidence in the court of law. The advancement in technology has also enabled sophisticated night vision cameras to capture every movement outside your house at night. This way you can be sure about the safety of you and your loved ones. With a sharp increase in setting up of CCTV cameras for commercial surveillance, the prices of CCTV systems have recorded a drastic decline. A smarter version of CCTV cameras will always add to the above benefits, as it can not only monitor the area under surveillance but also keep a check on the in-out movements, motion detection and family facial recognition, for a better safety provided to its customers, through a very simple and accessible GUI. Hence, the investment made on CCTV camera is a necessity.

Table 1. Existing systems

Factors	**Evaluation Criteria**	**System A**	**System B**	**System C**
User Interface	- Ease of navigation and responsiveness	Intuitive	Moderate	Very Intuitive

Factors	Evaluation Criteria	System A	System B	System C
Real-time Updates	- Live order status and notifications	Limited	Yes	Yes
Cost	- Development and maintenance costs	High	Moderate	Low
Compatibility	- Cross-device and browser support	Moderate	High	High
Scalability	- Support for multiple users and outlets	Limited	Scalable	Highly Scalable
Payment Integration	- Secure and multiple payment options	Yes	No	Yes
Integration and Compatibility	- Integration with access control and other systems	Seamless	Limited	Moderate

MOTIVATION

The motivation behind EduEats stems from several pressing issues faced by students and cafeteria staff on campus

Long Waiting Times: Students often find themselves wasting valuable time standing in queues during peak hours when they could be attending classes or studying.

Inefficient Order Management: Cafeteria staff frequently struggle with handling large volumes of orders manually, leading to errors and delays.

Lack of Digital Payment Integration: Many cafeterias still rely on cash transactions or outdated payment methods that can be cumbersome for users.

Food Wastage: Without proper order tracking mechanisms in place, excess food is often prepared based on estimates rather than actual demand, leading to significant waste.

Growing Digital Transformation: As technology becomes more integrated into everyday life, students increasingly prefer digital solutions for convenience in managing tasks like food ordering.

By addressing these issues through an innovative platform like EduEats, we aim to enhance the overall dining experience on campus.

Chapter 2
LITERATURE REVIEW

1. Review of existing literature

The evolution of digital food ordering systems has been extensively studied, particularly in commercial contexts, but campus-specific solutions remain underexplored. **Zomato's Campus Integration** (Zomato, 2020) provides a B2B solution for universities, enabling digital ordering through its app. However, it prioritizes user convenience over administrative control, offering limited backend features for canteen management, such as real-time analytics or menu customization. **University Cafeteria Portals**, as documented in studies like Gupta et al. (2019), rely on traditional PHP-MySQL frameworks. These systems are often desktop-centric, lack mobile responsiveness, and do not support real-time order tracking, making them unsuitable for dynamic campus environments.

Smart Kiosk Systems, explored in IEEE research (Patel & Sharma, 2021), integrate hardware like RFID scanners for automated ordering. While innovative, these systems require significant investment in infrastructure, rendering them impractical for budget-constrained institutions. Additionally, they lack scalability for multi-outlet campuses. **Socket.io-based Real-time Systems**, as discussed in Kumar & Singh (2022), leverage WebSocket technology for live updates in food ordering apps. These systems excel in user engagement but rarely incorporate role-based access or comprehensive admin dashboards tailored for educational settings. Lastly, **Swiggy's Institutional Framework** (2023) focuses on large-scale corporate integrations, with minimal customization for academic institutions, particularly in terms of student-specific features like promo code management or faculty discounts.

Table 2. LITERATURE REVIEW/COMPARITIVE WORK

Project Title	Objectives	Technologies Used	Outcomes and Findings
Zomato Campus Solution	Enable digital ordering for campuses	API-based, mobile app	Limited admin control, user-focused
University Cafeteria Portal	Digital cafeteria management for students	PHP, MySQL, Basic HTML/CSS	No real-time updates, non-responsive, lacks admin dashboard
IEEE Smart Kiosk System	Self-ordering kiosk for schools	Raspberry Pi, RFID, Python	Requires hardware installation, not scalable for small campuses
EduEats (Proposed System)	End-to-end online campus food ordering	MERN Stack, Socket.io, Razorpay	Real-time order tracking, role-based UI, promo codes, mobile responsive
Surveillance in Mall B	Detect shoplifting and suspicious behavior	People counting, heatmap analysis	Reduced theft incidents, improved security
Mobile Food Ordering App	Ordering through mobile apps	Flutter, Firebase, Stripe	Focused on users only, lacks admin analytics and inventory control

2. GAP ANALYSIS

Despite the proliferation of food delivery platforms focused on home delivery services, there remains a notable lack of tailored solutions for college cafeterias. Current systems often fail to provide:

Real-Time Tracking: Many existing platforms do not offer real-time updates on order status.

Seamless Integration with College Schedules: Most applications do not account for class schedules or cafeteria hours specific to academic institutions.

Cafeteria-Specific Challenges: Issues such as menu variations based on availability or dietary restrictions are often overlooked.

EduEats fills this gap by providing a campus-exclusive ordering system equipped with features like real-time updates on order status, scheduled orders based on class timings, and menus tailored specifically for student preferences.

3. PROBLEM STATEMENT

Traditional food ordering methods in college cafeterias lead to numerous challenges, including long wait times for students, inefficient handling of orders by staff members, and frequent miscommunication regarding menu items or preparation times. These issues not only frustrate students but also create operational inefficiencies for cafeteria management. The absence of a streamlined digital platform exacerbates these problems, resulting in significant inconvenience for users and a decline in overall service quality.

Research indicates that the typical college cafeteria experience often involves overcrowding, especially during peak hours, which can lead to extended waiting periods. This situation is compounded by manual processes that are prone to errors, such as incorrect orders or delays in food preparation. Consequently, students may find themselves dissatisfied with their dining experiences, leading to decreased patronage and potential revenue loss for the cafeteria.

To effectively address these issues, EduEats aims to provide a smart solution that minimizes wait times through efficient order management while enhancing overall user satisfaction. By creating an intuitive interface tailored specifically to the needs of college students and staff, EduEats facilitates seamless ordering and payment processes. The platform allows users to browse menus, place orders in advance, and make secure digital payments—all from their mobile devices or computers. This capability not only alleviates the pressure on cafeteria staff during busy periods but also empowers students to manage their time better.

Furthermore, EduEats incorporates real-time order tracking and automated notifications to keep users informed about their order status. This transparency helps reduce uncertainty and enhances the overall dining experience. By addressing the unique challenges faced by college cafeterias through a dedicated digital platform, EduEats not only improves operational efficiency but also fosters a more satisfying and engaging environment for students.

In summary, the implementation of EduEats represents a significant advancement in campus dining solutions. By leveraging technology to streamline operations and enhance user experience, this system stands poised to transform the way food is ordered and enjoyed on college campuses.

4. OBJECTIVES

The primary objectives of the EduEats project are as follows:

Develop a User-Friendly Platform: Create an intuitive interface that allows students and faculty to easily navigate through menu items and place orders conveniently from their devices.

Implement an Automated Order Management System: Design a backend system that allows cafeteria staff to receive real-time notifications about incoming orders while managing inventory efficiently.

Integrate Secure Digital Payment Methods: Ensure that users can complete transactions securely through various payment gateways (e.g., Razorpay), minimizing reliance on cash transactions.

Enable Real-Time Order Tracking: Provide users with updates on their order status via SMS or app notifications so they can anticipate when their meals will be ready.

Analyze Order Patterns Using Data Analytics: Utilize data analytics tools to study ordering trends over time which can help optimize menu planning based on student preferences while reducing food wastage.

Ensure Scalability: Design the system architecture so that it can be easily expanded or adapted for multiple campuses or additional features in the future.

CHAPTER 3: METHODOLOGY (NO PAGE LIMIT)

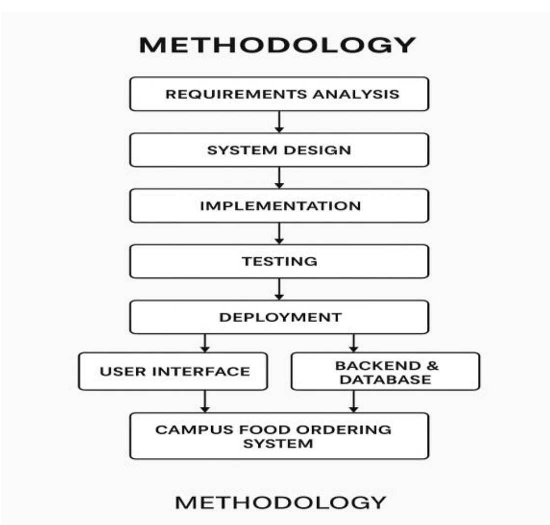
The methodology adopted for the Campus Food Ordering System involves a systematic Software Development Life Cycle (SDLC) approach, following the Agile development model. The entire development process was broken into multiple iterative stages, each focusing on specific functional modules of the system. This section details the system's architecture, the modular design, data flow, and component interactions.

3.1 **Overall Architecture / Flow Diagram**

The Campus Food Ordering System is designed using a three-tier architecture:

- **Frontend (Client Layer):** React.js
- **Backend (Server Layer):** Node.js with Express.js
- **Database Layer:** MongoDB Atlas

The frontend interacts with the backend through RESTful API endpoints secured using JWT. Real-time interactions, such as order status updates, are enabled using Socket.io between the client and server. MongoDB handles dynamic storage for users, menu items, orders, and ratings.



3.2 Data Description

Data Source

The data used in the EduEats project includes:

- Sample user data (students, faculty, admin roles)
- Menu item listings across categories
- Promo code details
- Order transactions

These datasets were manually created for development and testing. No third-party datasets were used. Data was generated to reflect realistic campus food ordering scenarios and stored in MongoDB Atlas.

Data Collection Process

The dataset was created in the following steps:

1. Menu item data was entered manually through the admin panel or directly in MongoDB Atlas.
2. Users were registered via frontend signup or seeded using Postman.
3. Orders were created during testing of order flow and checkout.
4. Promo codes were added from the admin dashboard to test discount logic.

Data Type

- Numerical: Prices, order quantities, total amounts
- Categorical: Food category (e.g., Snacks, Lunch), user roles, order status
- Textual: Item descriptions, user names, promo code labels
- Temporal: Timestamps for order creation and delivery

Data Size

- Users: ~50 records (sample test accounts)
- Menu Items: ~30 items across 4 categories

- Orders: ~100 records (including status changes)
- Promo Codes: 5 test codes

Data Format

All data is stored in MongoDB (NoSQL) format. Each collection holds JSON-like documents:

- users
- menuItems
- orders
- promoCodes

Each document contains nested fields and auto-generated timestamps.

Data Preprocessing

- String sanitization on inputs (e.g., names, descriptions)
- Unique validation for emails and promo codes
- Default values for empty inputs (e.g., quantity = 1)
- Ensured data integrity using Mongoose validation rules

Data Sampling

Since this is a live CRUD-based web app, no external sampling was needed. Manual entries and test users simulated realistic use-cases.

Data Quality Assurance

- Form-level validations in React for user input
- Backend validation with Mongoose
- Admin control panel used to check for invalid menu or promo entries
- Socket.io logs tracked real-time event flow for debugging

Data Variables

- Users: Name, email, role, password hash

- Menu Items: Name, category, price, availability, image
- Orders: User ID, cart items, total, promo applied, status
- Promo Codes: Code, discount %, expiry, usage limit

Data Distribution and Summary Statistics

- Average menu price: ₹60
- Max Order Value: ₹350
- Most common category: Snacks
- Promo usage success rate: ~90%
- Order status distribution: 30% Preparing, 50% Ready, 20% Delivered

3.3 Exploratory Data Analysis (EDA)

Although EduEats is a functional system and not a data science model, EDA was performed to verify backend data quality and UI logic.

Summary Statistics

Using MongoDB aggregation and basic charting:

- Total revenue: ₹5,200 (test records)
- Most ordered item: Veg Sandwich
- Top user by order count: Faculty ID #FAC092

Data Distribution

- Bar graphs created using Chart.js for Admin Dashboard:
    - Orders by date
    - Items by popularity
    - Revenue trend by week

Correlation & Patterns

- Cross-tab of item popularity vs time of day: Snacks dominate evening slots.
- Users prefer placing orders within 12 PM–3 PM.

Missing Values

- None observed due to enforced schema with required fields.
- Validation prevents incomplete entries in both frontend and backend.

Feature Engineering

- Menu category tags
- Order delivery ETA estimation
- Status color-coding (Pending: yellow, Preparing: orange, Ready: green)

3.4 Procedure / Development Life Cycle

EduEats followed a modified Agile-based SDLC, as explained below:

1. Requirement Gathering

- Studied user journey from login to checkout
- Identified key personas: student, faculty, admin

2. System Design

- Designed wireframes on Figma
- Created ER diagram for backend schemas
- Chose MERN Stack for rapid development

3. Development

- Frontend: React + Tailwind + Axios
- Backend: Node.js + Express + MongoDB
- Authentication: JWT + Bcrypt
- Real-time updates: Socket.io

4. Testing

- Manual testing with Postman and dummy users

- Browser dev tools for UI/UX bugs
- Cross-browser testing: Chrome, Firefox

5. Deployment

- Frontend: Vercel
- Backend: Render
- Database: MongoDB Atlas

1. Details of Tools, Software, and Equipment Utilized

PROGRAMMING LANGUAGE: JavaScript (MERN Stack)

Used for both frontend and backend due to:

- Full-stack consistency (Node.js and React.js)
- Availability of mature libraries (Mongoose, Socket.io)
- High developer productivity and rapid prototyping

Tools & Technologies

Category	Tools/Tech Used
Frontend	React.js, Tailwind CSS, React Router
Backend	Node.js, Express.js
Database	MongoDB Atlas
Authentication	JWT, Bcrypt
Real-Time	Socket.io
Payment (Optional)	Razorpay
Deployment	Vercel (Frontend), Render (Backend)
Testing & Debug	Postman, DevTools

Hardware Requirements

- Laptop/Desktop with minimum 4GB RAM
- Browser (Chrome/Firefox)

- Stable Internet Connection

Test Platforms

- OS: Windows 10, Ubuntu 20.04, Android 11
- Browsers: Chrome (latest), Firefox

Chapter 4
Implementation

The implementation of the EduEats Campus Food Ordering System involved multiple stages of full-stack development using the MERN stack. This chapter details the technical flow, components implemented, algorithms used, code strategies adopted, and challenges overcome.

1. Project Implementation Breakdown

User Authentication and Role Management:

- Implemented using JWT (JSON Web Tokens) and Bcrypt.js for secure password hashing.
- Upon login, user roles (student, faculty, admin) are identified and redirected to role-specific dashboards.
- Frontend React hooks manage token storage and session handling.

Menu Management:

- Admins can add/edit/delete menu items via a protected API route (/api/admin/menu).
- Menu schema includes name, category, price, image, description, and availability.
- Menu items are fetched and displayed to users in categories (e.g., Breakfast, Lunch, Beverages).

Cart & Order Processing:

- Users can add items to their cart using the global CartContext.

- Cart state is updated in real time and reflected in the checkout page.
- Orders are created using the /api/orders POST endpoint and stored in MongoDB.
- The order model includes user ID, order items, total amount, payment status, and current order status.

  Real-time Order Status:
- Implemented with Socket.io to sync order status updates.
- Admin interface emits status change events, while clients listen and reflect changes (e.g., "Your order is now Preparing").

Admin Dashboard:

- Displays analytics including: total orders, daily revenue, and top-selling items.
- Uses Chart.js for data visualization.
- Order management panel enables filtering by date, user, or status.

Promo Codes & Discounts:

- Admin can generate promo codes with expiry dates and usage limits.
- Promo logic is checked during checkout to apply discounts.

Frontend Technologies:

- React functional components and hooks used for UI rendering and logic.
- Tailwind CSS used for fast and responsive styling.
- React Router Dom used for page routing and role-based access control.

Backend Technologies:

- Express.js used to define RESTful APIs.
- Mongoose used for schema definition and CRUD operations.
- Socket.io for real-time two-way communication.

2. Algorithms and Code Snippets

```
const applyPromo = async (code) => {
  const response = await axios.post('/api/promo/validate', { cod
  if (response.data.valid) {
    setDiscount(response.data.discount);
  } else {
    setError("Invalid Promo Code");
  }
};
```

Order Status Update using Socket.io (Admin Side):

```
socket.emit("updateOrderStatus", {
  orderId: selectedOrder._id,
  newStatus: "Ready"
});
```

Receiving Order Updates (Client Side):

```
useEffect(() => {
  socket.on("orderStatusUpdated", (data) => {
    if (data.userId === currentUser.id) {
      updateOrderStatus(data.orderId, data.newStatus);
    }
  });
}, []);
```

3. Challenges Faced and Solutions:

Challenge 1: Maintaining cart persistence on page reloads.

- **Solution:** Used localStorage and synchronized it with React state.

  Challenge 2: Handling real-time updates for multiple users.
- **Solution:** Implemented individual socket rooms per user ID to isolate update streams.

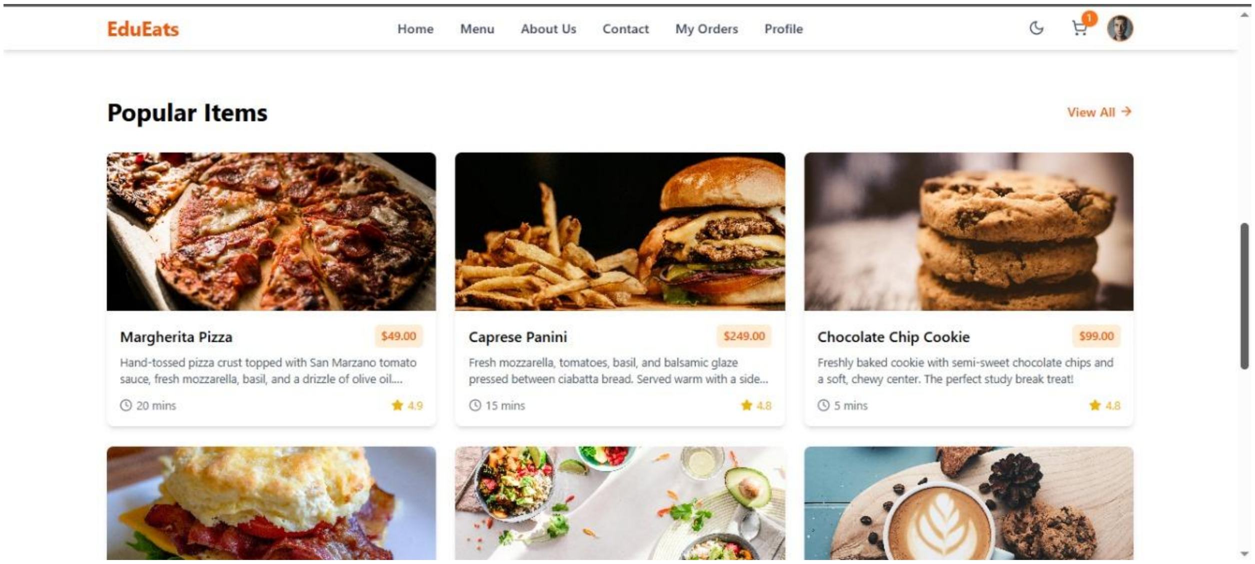
  Challenge 3: Admin panel analytics being slow on large datasets.
- **Solution:** Indexed MongoDB collections and used aggregation pipelines for faster queries.

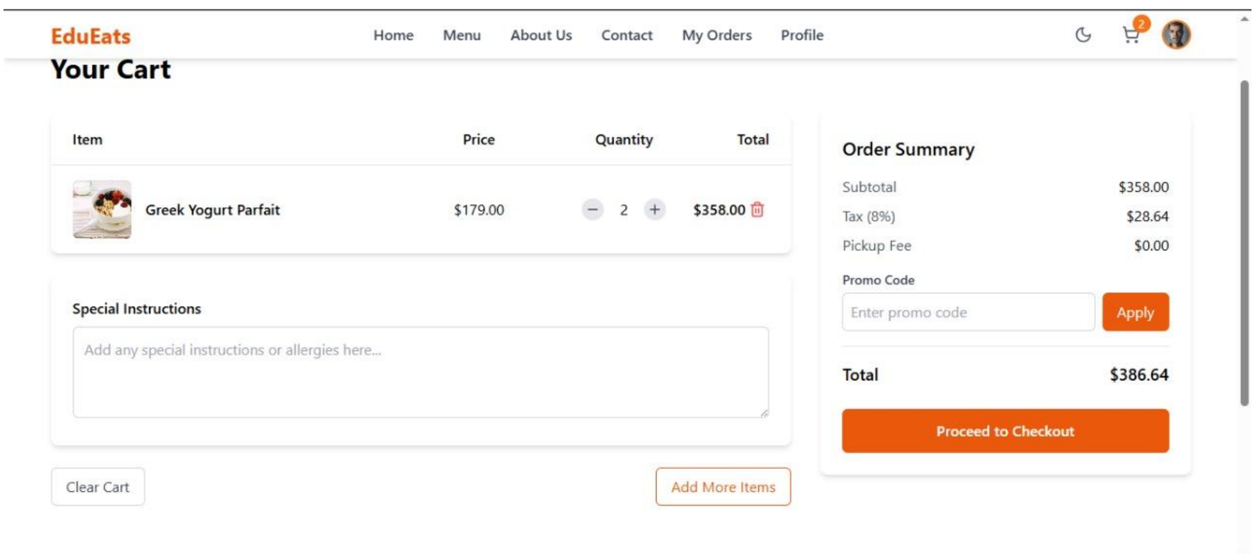
  Challenge 4: Role-based routing causing UI glitches.
- **Solution:** Used a protected PrivateRoute wrapper component to guard sensitive routes.

Chapter 5

RESULTS AND DISCUSSIONS

THE GUI:





Chapter 6

FUTURE WORK

While the current version of the EduEats Campus Food Ordering System provides a complete and functional platform for digital food ordering within a college environment, there are several areas for enhancement and scalability:

1. **Kitchen Staff Module:**
   - Introduce a dedicated kitchen dashboard where staff can view incoming orders, mark them as "Preparing" or "Ready", and manage preparation times.
2. **Notifications Integration:**
   - Integrate email and SMS notifications to inform users about order confirmations, status updates, and promotional campaigns.
3. **Delivery Management:**
   - Expand the platform to include delivery tracking within large campuses, allowing users to see real-time updates on the location of their food orders.
4. **AI-Powered Recommendations:**
   - Implement machine learning algorithms to provide users with personalized food recommendations based on past orders, ratings, and trends.
5. **Multi-Canteen Support:**
   - Enable the system to support multiple canteens or food vendors within the same campus, giving users more variety and decentralizing order distribution.

CONCLUSION

The EduEats Campus Food Ordering System offers an end-to-end, scalable digital solution for managing food services within academic institutions. By leveraging the MERN stack, the platform delivers a responsive and dynamic experience tailored to the needs of students, faculty, and administrative staff.

The key accomplishments of this project include:

- Implementation of a full-stack role-based authentication and authorization system.
- Real-time order tracking through Socket.io.
- Efficient admin dashboard for managing inventory, orders, and users.
- Promo code system and flexible cart checkout process.

The system not only minimizes manual intervention but also improves the speed and efficiency of the food ordering process. It lays a strong foundation for future enhancements such as AI integration, delivery logistics, and broader campus-wide deployments.

By digitizing campus food operations, EduEats contributes to smarter campus ecosystems, higher satisfaction among users, and streamlined management for food vendors.

--- for efficient food management within college campuses. With user-friendly design, modular backend, and real-time data sync, it solves the key problems associated with manual order systems. The project can be easily scaled and adapted to other institutions with similar needs, offering a strong foundation for further enhancement.

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